

01 - Summary

Title: UNAcore

Usage: Main PCB for UNA Watch

Modifications Record:

1.0)	Design Completed	17/01/24	Lewis Allison
1.1)	Engineering Change Note 1 (Layout Only)	29/01/24	Lewis Allison
1.2)	Engineering Change Note 2	02/02/24	Lewis Allison
1.3)	Engineering Change Note 3 (Layout Only)	05/02/24	Lewis Allison
2.0)	Engineering Change Note 4	10/04/25	Lewis Allison
2.1)	Engineering Change Note 5 (Layout Only)	18/04/25	Lewis Allison
2.2)	Engineering Change Note 6	08/08/25	Lewis Allison
3.0)	Engineering Change Note 7	10/03/26	Lewis Allison
3.1)	Engineering Change Note 8	27/04/26	Lewis Allison
3.2)	Engineering Change Note 9 (BOM Only)	05/05/26	Lewis Allison

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32-bit Cortex-M33 MCU



```
VCK
ENB
HCK
BUZZER_PWM
LCD_FRP
LCD_XFRP
```

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ECN 4:
U1 changed to STM32U595QII60.
X1 added (ECS-160-8-37-CWY-TR3).
C69, C70 added (GRM0335C1H8R0CA01D).
HSE CLK connected to MCU.
Added EMMC_EN net to enable EMMC power from MCU.
SENSORBODY_RST_L moved to pin PF13.
DEBUG_TX connected to PC4 and routed out to USB.
Added resistor option to route SWO instead of DEBUG_TX.
Connected ALTIMETER_EN and ALTIMETER_INT to pins PG0 and PF0.
```

ECN 7:
Connected DAM_DETECT to U1.K12.
Connected USB_FAULT_L to U1.D2.

Diagram illustrating the SWO pin configuration for the R24 NOPO module. The module has five pins (TP-1 to TP-5) with dimensions and connections as follows:

- TP-1: 1.0mm dia. SMT
- TP-2: 1.0mm dia. SMT
- TP-3: 1.0mm dia. SMT
- TP-4: 1.0mm dia. SMT
- TP-5: 1.0mm dia. SMT

The connections are:

- SWDIO <-> TP-1
- SWCLK <-> TP-2
- BOOT0 <-> TP-3
- RST_L <-> TP-4
- SWO <-> TP-5
- DEBUG_TX <-> TP-5

The SWO and DEBUG_TX signals are connected to the R24 NOPO module.

SENSOR1_INT2	PA0
SENSOR2_INT2	PA1
SENSOR2_TX_INT0	PA2
SENSOR2_RX_INT1	PA3
SW1	PD4
SW3	PA5
FUEL_GAUGE_ALRT	PC6
SENSORBODY_INT3	PG6
PWR_INT_L	PG7
SW4	PD8
SENSOR1_RX_INT0	PA9
SENSOR1_RX_INT1	PA10
SENSORBODY_INT0	PB11
SENSORBODY_INT1	PF12
TOUCH_INT_L	PC13
SENSORBODY_INT2	PF14
PWR_ON_V18_L	PD15

SDIO_CMD	EMMC_EN	<3>
SDIO_CK	SDIO_CK	<3>
SDIO_D0	SDIO_D0	<3>
SDIO_D1	SDIO_D1	<3>
SDIO_D2	SDIO_D2	<3>
SDIO_D3	SDIO_D3	<3>

```

HAPTIC_RST_N      >>> HAPTIC_RST_N <4>
BUZZER_EN1        >>> BUZZER_EN1  <4>
BUZZER_EN2        >>> BUZZER_EN2  <4>
BUZZER_PWM        >>> BUZZER_PWM  <4>
ALTIMETER_EN      >>> ALTIMETER_EN <4>

```

MAIN_SDA	MAIN_SDA	<4,5,6,7>
MAIN_SCL	MAIN_SCL	<4,5,6,7>
FUEL_GAUGE_ALRT	FUEL_GAUGE_ALRT	<5>
RST_L	RST_L	<2,5,6>
PWR_INT_L	PWR_INT_L	<5,6>
PWR_ON_1V8_L	PWR_ON 1V8_L	<5>

USB_D_P USB_D_P <6>
 USB_D_N USB_D_N <6>
 DAM_DETECT DAM_DETECT <6>
 USB_FAULT_L USB_FAULT_L <6>

LCD_QSPI_SCK_VST	>>>	LCD_QSPI_SCK_VST	<>
LCD_QSPI_IO0_XRST	>>>	LCD_QSPI_IO0_XRST	<>
LCD_QSPI_IO1_HST	>>>	LCD_QSPI_IO1_HST	<>
LCD_QSPI_IO2_ENB	>>>	LCD_QSPI_IO2_ENB	<>
LCD_QSPI_IO3_HCK	>>>	LCD_QSPI_IO3_HCK	<>
LCD_QSPI_CS_VCK	>>>	LCD_QSPI_CS_VCK	<>

```

LCD_XFRP          >>> LCD_XFRP  <7>
LCD_R1            >>> LCD_R1   <7>
LCD_R2            >>> LCD_R2   <7>
LCD_G1            >>> LCD_G1   <7>
LCD_G2            >>> LCD_G2   <7>
LCD_B1            >>> LCD_B1   <7>
LCD_B2            >>> LCD_B2   <7>

TOUCH_INT_L       >>> TOUCH_INT_L  <7>
TOUCH_RST_L       >>> TOUCH_RST_L  <7>
BACKLIGHT_ON      >>> BACKLIGHT_ON  <7>

```

SENSOR1_SCL	>>>	SENSOR1_SCL	<<<
SENSOR1_SDA	>>>	SENSOR1_SDA	<<<
SENSOR1_SCK	>>>	SENSOR1_SCK	<<<
SENSOR1_MISO	>>>	SENSOR1_MISO	<<<
SENSOR1_MOSI	>>>	SENSOR1_MOSI	<<<
SENSOR1_CS	>>>	SENSOR1_CS	<<<
SENSOR1_TX_INT0	>>>	SENSOR1_TX_INT0	<<<
SENSOR1_RX_INT1	>>>	SENSOR1_RX_INT1	<<<
SENSOR1_INT2	>>>	SENSOR1_INT2	<<<
SENSOR1_RST_L	>>>	SENSOR1_RST_L	<<<

SENSOR2_SCL	>>	SENSOR2_SCL	<>
SENSOR2_SDA	>>	SENSOR2_SDA	<>
SENSOR2_SCK	>>	SENSOR2_SCK	<>
SENSOR2_MISO	>>	SENSOR2_MISO	<>
SENSOR2_MOSI	>>	SENSOR2_MOSI	<>
SENSOR2_CS	>>	SENSOR2_CS	<>
SENSOR2_TX_INT0	>>	SENSOR2_TX_INT0	<>
SENSOR2_RX_INT1	>>	SENSOR2_RX_INT1	<>
SENSOR2_INT2	>>	SENSOR2_INT2	<>
SENSOR2_RST_L	>>	SENSOR2_RST_L	<>

<u>SENSORBODY_SCL</u>	SENSORBODY_SCL	<A7>
<u>SENSORBODY_SDA</u>	SENSORBODY_SDA	<A7>
<u>SENSORBODY_SCK</u>	SENSORBODY_SCK	<A7>
<u>SENSORBODY_MISO</u>	SENSORBODY_MISO	<A7>
<u>SENSORBODY_MOSI</u>	SENSORBODY_MOSI	<A7>
<u>SENSORBODY_CS</u>	SENSORBODY_CS	<A7>
<u>SENSORBODY_INT0</u>	SENSORBODY_INT0	<A7>
<u>SENSORBODY_INT1</u>	SENSORBODY_INT1	<A7>
<u>SENSORBODY_INT2</u>	SENSORBODY_INT2	<A7>
<u>SENSORBODY_INT3</u>	SENSORBODY_INT3	<A7>
<u>SENSORBODY_RST_L</u>	SENSORBODY_RST_L	<A7>

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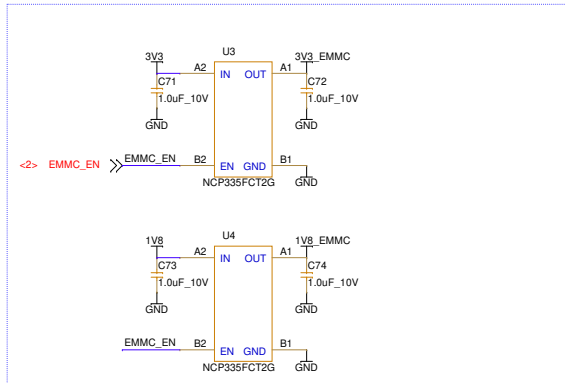
Title: UNACore

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03 - eMMC

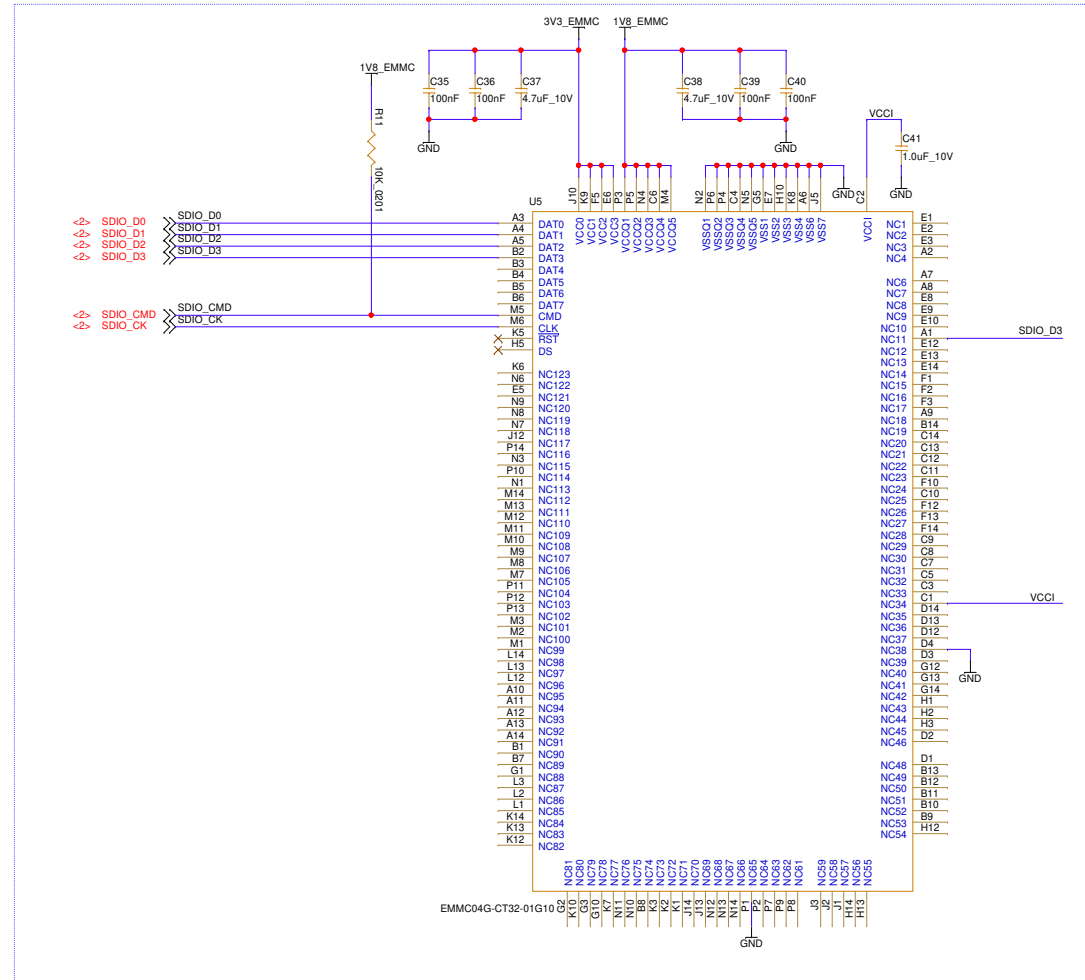
Load Switches to Reduce Power Consumption



ECN 4:
U3 & U4 added (NCP335FCT2G).
C71, C72, C73, C74 added (GRM033R61A105ME15D).
R22 & R23 added (No Pop R0201).

ECN 7:
Removed R22, R23.

4GB eMMC



ECN 2:
C42 changed to GRM033R61A105ME15D (1uF_10V)

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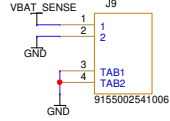
	Rev
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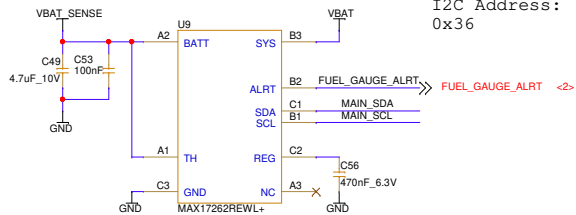
05 - PMIC

Battery Connector

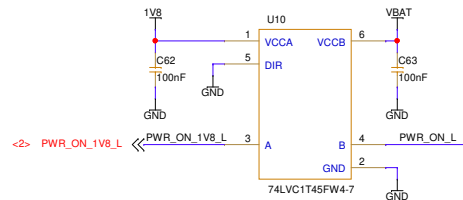


ECN 4:
J9 replaced by 9155002541006.
J10 removed.
U12, R14, R15 & C65 removed
(Protection circuit now on battery).

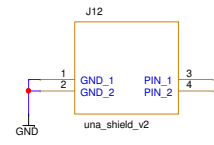
Power Gauge



Level Translator for Hold Button Detection

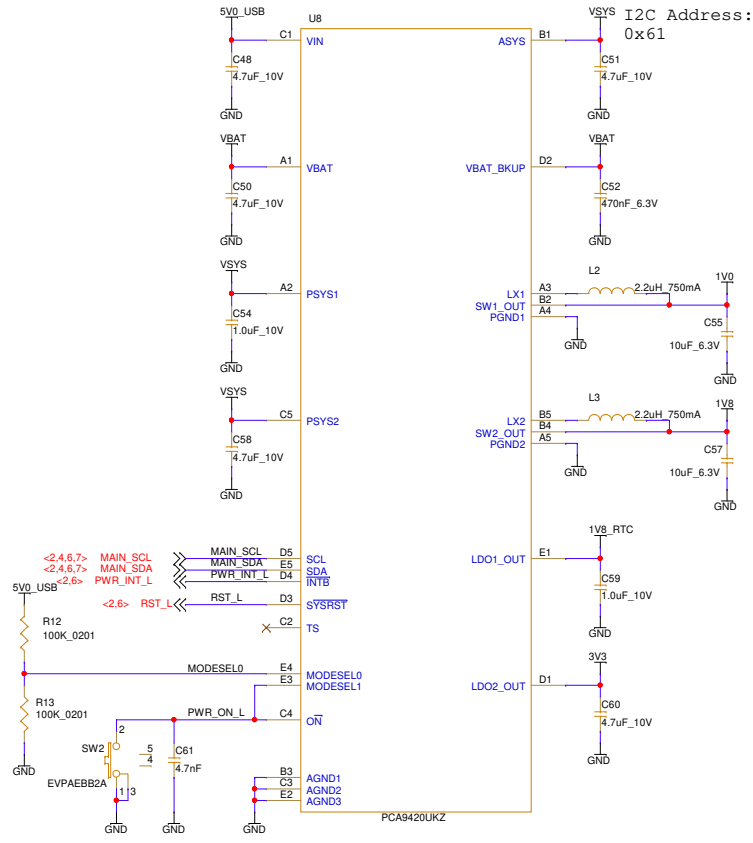


Board Shield



ECN 2:
U12 changed to SGM41100A-450009YUDT6G/TR
R15 added (NOPOP_0402)
C54, C59, C65 changed to GRM033R61A105ME15D (1uF_10V)

PMIC

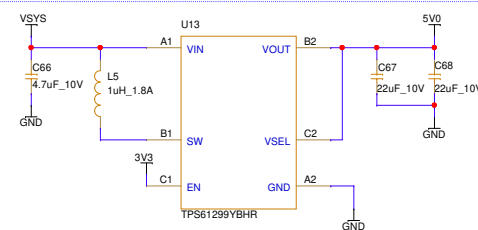


ECN 4:
TP12 removed (BAT_N no longer exists).
TP33 & TP34 added.

ECN 6:
TP36 added.

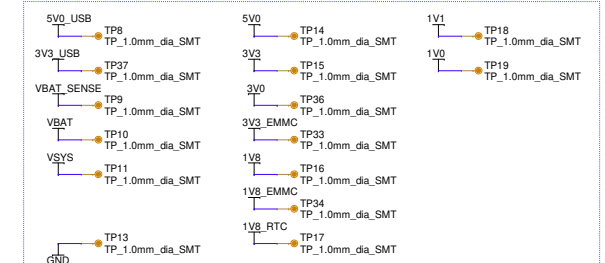
ECN 7:
TP37 added.

5V Boost Supply



Note:
Each 22 uF capacitor drops to
3.4 uF effective C under 5V bias.

Power Rail Test Points



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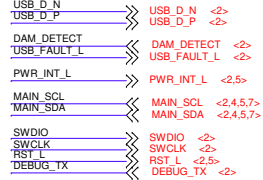
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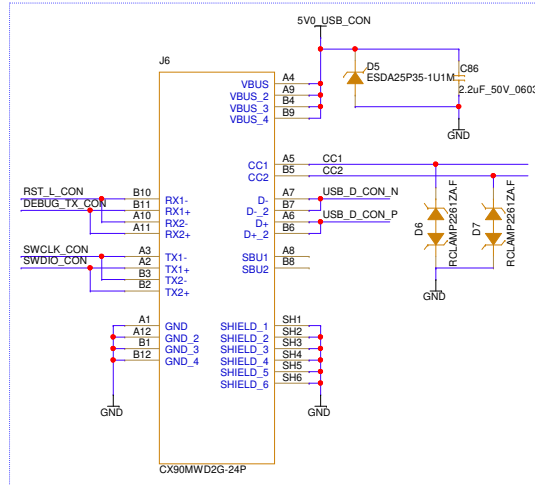
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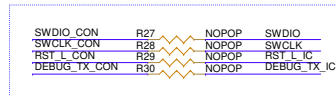
06 - USB



USB-C IPX8 Connector



Option to Bypass Switches (Internal Use Only)



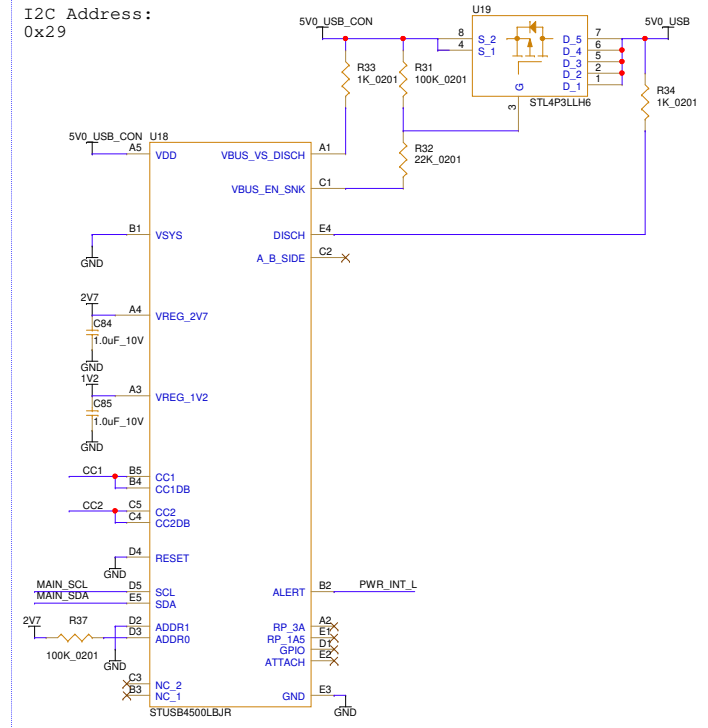
ECN 4:
 J6 replaced by CX90MWD2G-24P.
 Added U14 (TMUX1511RSVR)
 Added J11 (MS0-3630242)
 Added C75 (100nF)
 Added R18 (10K)
 Added U15 (NCP335FCT2G)
 Added C76 & C77 (1uF)

ECN 7:
 Removed L4
 Removed R16, R17
 Removed R18
 Removed U11
 Removed U14
 Removed U15

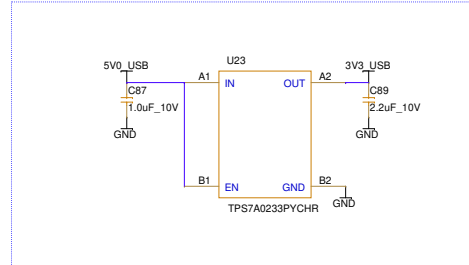
Added C82 (100nF_50V)
 Added C83, C84, C85, C87 (1uF)
 Added C86 (2.2uF_50V_0603)
 Added C89 (2.2uF)
 Added D1, D2, D3, D4 (RCLAMP2261ZA.F)
 Added D5 (ESDA25P35-1U1M)
 Added R27, R28, R29, R30 (NOPOP)
 Added R31, R37 (100K)
 Added R32 (22K)
 Added R33, R34 (1K)
 Added R39, R40 (200R)
 Added U18 (STUSB4500LBJR)
 Added U19 (STL4P3LLH6)
 Added U20 (PTUSB251UCX)
 Added U21, U22 (NX20P0407UKZ)

USB-C Controller, 5V Only

I2C Address:
 0x29



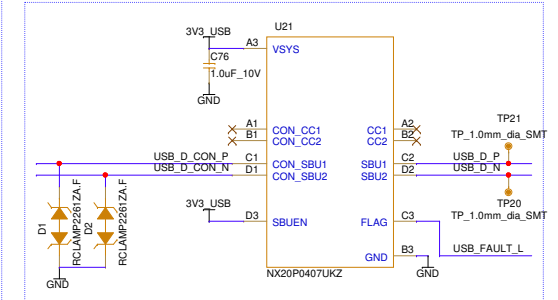
3V3 LDO (Active only with USB Cable)



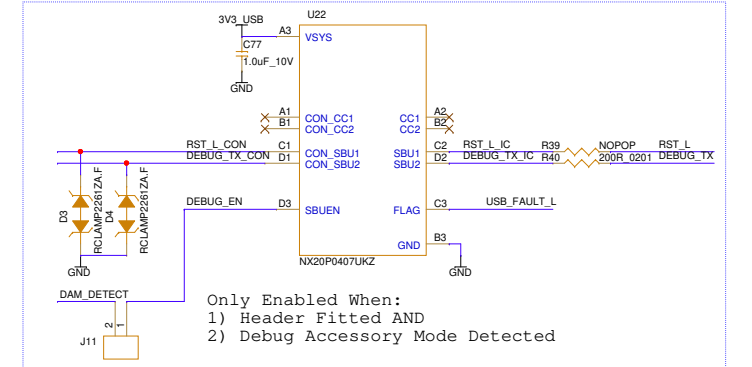
ECN 8:
 Removed U20
 Removed C82
 Removed C83
 CC1, CC2 connected directly from J6 to U18
 D6, D7 added (RCLAMP2261ZA.F)

ECN 9:
 R39 changed to NOPOP (DO NOT FIT)

DP & DN Short to VBUS Protection



RST_L & DEBUG_TX Short to VBUS Protection



Only Enabled When:
 1) Header Fitted AND
 2) Debug Accessory Mode Detected

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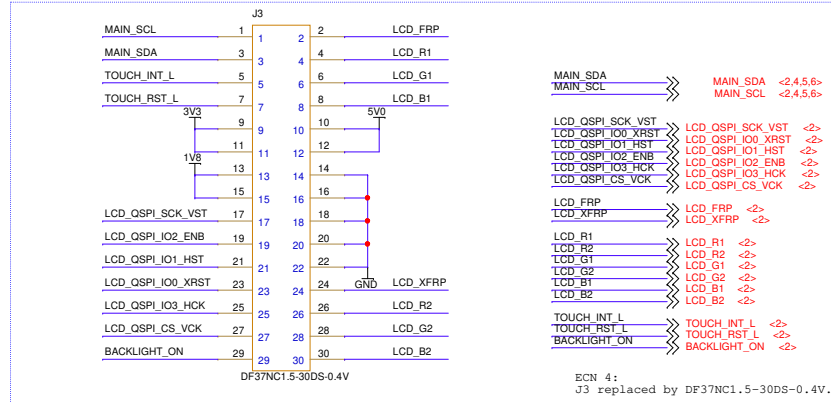
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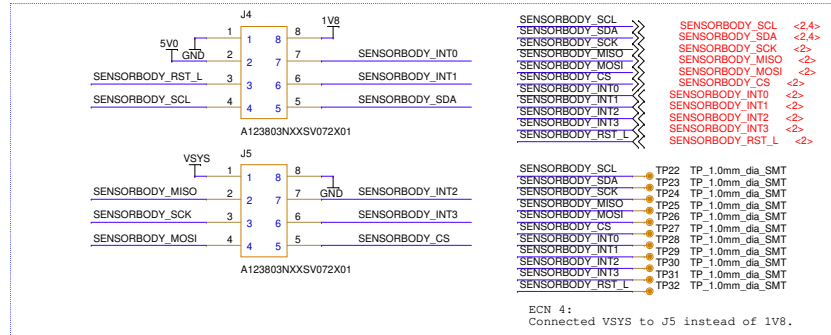
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06 - Connectors to Modules

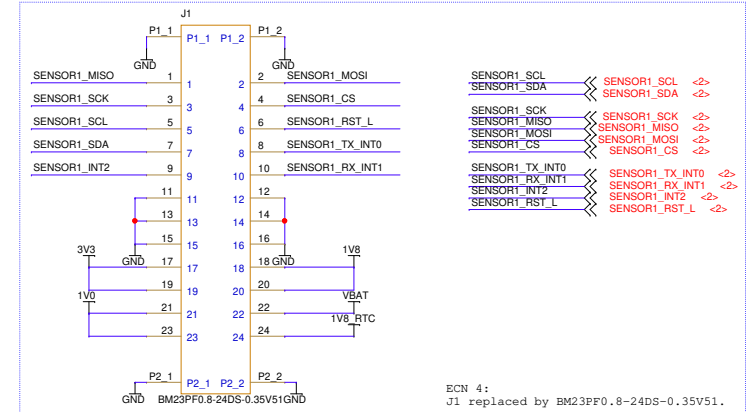
Display Module Interface



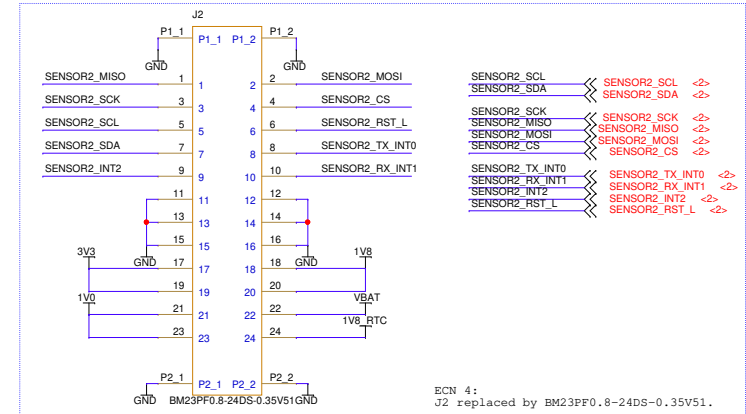
Body Sensor Module Interface



Sensor 1 Module Interface



Sensor 2 Module Interface



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